MicroBioAgent: an AI agent for automated microbiome analysis

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Abstract

Microbiome studies require analytical choices that depend on the biological question, available assays, and intermediate findings. Translating these choices into executable analyses requires connecting heterogeneous methods, software environments, reference resources, and intermediate outputs. Here we present MicroBioAgent, a domain agent that translates biological questions into study-specific analyses across 16S amplicon, reference-based shotgun, and assembly-based metagenomic workflows. MicroBioAgent uses microbiome-specific skills to select and execute analytical methods in user-controlled computational environments. It evaluates intermediate results to determine subsequent analyses in line with the biological question. We evaluated MicroBioAgent on three published datasets through amplicon and shotgun analysis of tomato rhizosphere communities, taxonomic and functional profiling with classifier training in a colorectal cancer cohort, and metagenomic assembly and MAG generation from a model polyethylene degradation study, followed by taxonomic profiling and annotation of genes associated with alkane and oxidized-derivative degradation in focal MAGs. The reanalyses identified biological patterns consistent with the original studies, with differences in statistical support and taxon detection across analytical methods and reference databases. The system also retains the analysis context across long-running operations, allowing pending work to be revised and execution to continue after tool completion. These applications demonstrate that MicroBioAgent can carry out upstream processing and downstream analysis across different microbiome study designs without requiring a fixed end-to-end workflow template.

1 Introduction

Microbiome studies seek to explain how microbial communities vary with host phenotype, environmental conditions, and functional potential. The analytical route depends on the question and the available data: 16S rRNA gene amplicons characterize community diversity and taxonomy, reference-based shotgun profiling estimates species and functional composition, and assembly-based analysis constructs metagenome-assembled genomes (MAGs)

for taxonomic and functional characterization [1]. A study may combine these routes or extend an analysis after an intermediate finding motivates a new comparison. Researchers must therefore connect biological objectives to method selection, compatible inputs, reference resources, and statistical interpretation. This work requires both domain knowledge and computational proficiency, creating barriers for researchers without specialist bioinformatics training [2, 3].

Workflow managers such as Snakemake and Nextflow provide a foundation for building reproducible and scalable analysis pipelines [4, 5]. They organize computational dependencies and coordinate execution of a specified workflow. Applying a pipeline to a biological study still requires researchers to select an appropriate analysis path, configure methods and resources, and interpret the resulting outputs. Extending a study to another assay or adding a comparison prompted by an intermediate finding may require changes to that path. Translating a research question into the workflow, and revising it as the analysis develops, therefore remains a substantial part of the researcher's task.

Large language model agents offer a means of connecting natural-language objectives to computational actions. AutoBA demonstrated planning, code generation, execution, and repair for multi-omics analyses, while Biomni combined planning and tool use across biomedical tasks [6, 7]. Other systems have explored broader bioinformatics execution and human supervision of multi-step analyses [8–10]. These developments suggest that agents can help compose the analytical path as well as execute its operations. In microbiome research, this approach requires domain knowledge that connects method selection to assay type, software and database requirements, expected outputs, and downstream interpretation. A central design question is how to organize that knowledge so that an agent can assemble study-specific analyses across complementary microbiome workflows.

Here we present MicroBioAgent, a domain agent that translates microbiome research questions into inspectable, study-specific analyses executed in user-controlled computational environments. It represents analytical methods as modular domain procedures spanning 16S amplicon profiling, reference-based shotgun analysis, and assembly-based MAG construction. The agent selects procedures suited to the research question and executes them using configured software and reference databases. Before execution, it develops an analysis plan for the study. During execution, it examines intermediate results to determine whether the remaining steps need to be revised. MicroBioAgent also supports asynchronous human-in-the-loop interaction: users can provide feedback while a long-running operation continues, allowing the agent to revise pending analyses and resume execution automatically after the operation completes.

MetaClaw is another AI agent designed for microbiome analysis. It combines predefined, deterministic upstream workflows on FlowHub with

customizable downstream analyses in an OpenClaw skill container. A single YAML pipeline registry links the two layers [11]. MicroBioAgent composes upstream processing and downstream analyses from modular skills without requiring a predefined end-to-end workflow template. Both stages can run in a user-controlled computational environment, allowing researchers to retain control over their sequencing data without uploading raw reads to an external analysis platform. We reanalysed three published datasets comprising 230 sequencing libraries from plant, human, and environmental microbiomes to evaluate whether MicroBioAgent could complete diverse analytical tasks and recover biological patterns reported in the original studies despite differences in methods and reference databases. To assess robustness, we designed 27 independent sessions spanning three language models, three levels of prompt detail, and three repeats per combination. We also designed component ablations to examine how domain skills, execution guidance, and the runtime extension contribute to task completion, method adherence, and token consumption. A separate runtime comparison tested whether user feedback could update the plan during an ongoing analysis and whether execution resumed automatically after tool completion. Together, this work aims to enable researchers without specialist bioinformatics training to carry out end-to-end microbiome analysis while preserving flexibility in method selection and user control during execution.

2 Results

2.1 Composing study-specific analyses across microbiome workflows

MicroBioAgent is an AI-agent framework that integrates flexible composition of upstream and downstream microbiome analyses, local execution, and asynchronous human-in-the-loop interaction (figure 1). Given a research question expressed in natural language, it develops an analysis plan and selects microbiome skills to carry it out. These skills provide method-specific guidance that the agent draws on when choosing analytical approaches, preparing commands, and interpreting outputs. As each step completes, the agent examines its results and decides whether the pending plan needs adjustment.

MicroBioAgent executes upstream processing and downstream analyses in a user-controlled local computational environment, without requiring raw sequencing data to be uploaded to an external analysis platform. To support these analyses, the system manages analytical tools and reference databases through separate configuration files. The tool configuration specifies available software and its associated execution environments, while the database configuration records reference resources and their local storage paths. Users can modify these configurations to add or replace tools and reference databases

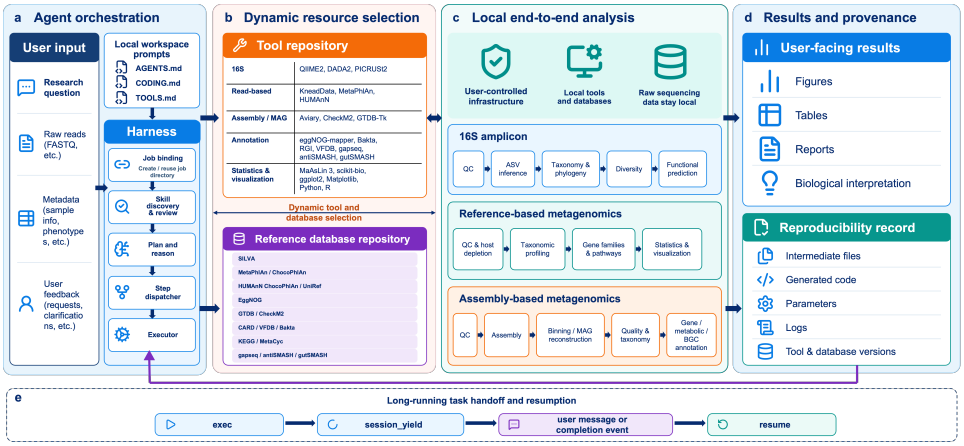


Figure 1: MicroBioAgent architecture and operating model. A user request is bound to an analysis job, from which the agent plans, selects skills, and resolves local runtimes and reference resources for 16S amplicon, reference-based shotgun, and assembly-based metagenomic analyses. Intermediate artifacts, commands, parameters, and logs remain associated with the job. During long-running computation, a user message can update pending work, whereas command completion triggers output verification and continuation in the same session.

according to their research needs.

For upstream analysis, MicroBioAgent supports 16S amplicon processing, reference-based metagenomic profiling, and assembly-based metagenomic reconstruction (table 1). Its amplicon capabilities include sequence quality assessment, denoising, amplicon sequence variant inference, taxonomic classification, and phylogenetic reconstruction. For shotgun data, it supports read quality control and host-read removal, taxonomic and functional profiling, and gene-catalogue construction. Assembly-based capabilities include metagenomic assembly and binning, MAG quality assessment, dereplication, taxonomic assignment, and abundance estimation.

MicroBioAgent supports a range of downstream analyses, such as comparisons of microbial communities and characterization of the functions of individual organisms. These include comparisons of diversity and taxonomic or functional abundances between groups, covariate association testing, functional enrichment, association network analysis, and supervised classification. For genes and recovered genomes, available analyses include functional and pathway annotation, antimicrobial-resistance and virulence-gene annotation, metabolic reconstruction, and identification of primary metabolic and secondary-metabolite biosynthetic gene clusters. For visualization, MicroBioAgent supports ordination plots of community differences and heatmaps of taxonomic abundances or functional profiles, with customization to highlight

the comparisons of interest.

During long-running computations, such as MetaPhlAn or HUMAnN profiling, MicroBioAgent keeps the conversation available for user interaction while the analysis runs in the background. Users can provide additional instructions or revise the pending analysis plan without waiting for the running task to finish. This asynchronous human-in-the-loop interaction allows the agent to incorporate feedback during execution and automatically resume the analysis under the updated plan after checking the completed task's outcome. MicroBioAgent thus supports end-to-end microbiome analysis through flexible workflow composition and local execution, while allowing users to guide ongoing analyses.

Table 1: MicroBioAgent analytical capabilities, grouped by scientific task. Functional predictions and genome annotations describe inferred biological potential.

Task category	Supported analyses
Amplicon characterization	Read import and quality summaries, denoising and ASV inference, taxonomy, phylogeny, predicted functional profiles
Shotgun profiling	Read quality control and host depletion, species abundances, gene-family and pathway profiles, nonredundant gene catalogues
MAG characterization	Assembly and binning, quality assessment. Dereplication, taxonomic assignment, abundance estimation, genome annotation
Community comparisons	Alpha and beta diversity, ordination, group comparisons, differential abundance, covariate association testing
Functional interpretation	Gene and protein annotation, orthology, pathway, and ontology mapping, functional enrichment, resistance and virulence gene annotation
Metabolic characterization	Metabolic pathway and model reconstruction, primary metabolic gene clusters, secondary-metabolite biosynthetic gene clusters
Study-specific analysis	Custom statistical scripts, association networks, supervised classification, figures, result tables, and analytical reports
Data and resource access	Public sequencing metadata and reference-protein retrieval, local reference resources and online annotation queries

2.2 Study-specific analyses across plant, human, and environmental microbiomes

We evaluated MicroBioAgent on three published datasets comprising 230 sequencing libraries from plant, human, and environmental microbiomes (Supplementary Table S1). These case studies assessed the system's ability to

complete the requested analyses and recover biological patterns reported in the source studies. Because software versions, reference databases, and some statistical procedures differed, exact computational reproduction was not the evaluation objective. Instead, we assessed completion of the requested analyses, validity of the resulting outputs, and recovery of prespecified biological signals.

The three cases were selected to test MicroBioAgent on distinct combinations of data types and analytical tasks. The tomato rhizosphere case combined 60 16S amplicon libraries with 36 matched shotgun libraries to examine differences in microbial diversity and taxonomic composition associated with pathogen inoculation and cultivar resistance. The colorectal cancer case used 128 faecal shotgun metagenomes from 74 patients and 54 controls to compare species and functional abundances and train classifiers based on gene-family and KEGG Orthology profiles. The third case used six shotgun metagenomes from soil-derived enrichment cultures grown on low-molecular-weight polyethylene or alkane substrates to generate MAGs through assembly and binning, dereplicate them, characterize their taxonomic composition, and annotate degradation-associated genes in focal genomes.

2.2.1 A tomato rhizosphere bacterial-wilt resistance study

We applied MicroBioAgent to a published study of tomato rhizosphere microbiomes associated with resistance to bacterial wilt [12]. The input comprised 60 16S V3–V4 amplicon libraries and 36 matched shotgun libraries from two susceptible cultivars (A57 and HG70), a resistant parent (HG64), and a resistant F1 hybrid (HF12). The amplicon data covered five sampling conditions with three biological replicates per cultivar: before inoculation (T1), and inoculated and uninoculated plants at day 5 (T2 and T2C) and day 10 (T3 and T3C). The matched shotgun libraries covered T1, T2C, and T2. Starting from sequencing reads, MicroBioAgent performed quality control, DADA2 denoising and taxonomic classification against the SILVA database for the amplicon data, and quality control, host-read removal and MetaPhlAn3 taxonomic profiling for the shotgun data. The resulting abundance tables were used for downstream ecological and statistical analyses.

For the 16S amplicon data, MicroBioAgent examined how pathogen inoculation and cultivar resistance were reflected in microbial diversity and community composition. Following *Ralstonia* inoculation, Shannon diversity declined in susceptible cultivars and remained higher in resistant cultivars at both day 5 ($P = 0.002$) and day 10 ($P = 0.004$, figure 2a). Community composition also differed between inoculated and uninoculated samples, which separated along the first weighted-UniFrac PCoA axis (56.3% of variance). Resistant and susceptible samples showed a less pronounced separation along the second axis (12.2%), with overlap between the two groups (figure 2b). Genus-level profiles linked these community changes to an expansion of

Ralstonia, most pronounced in the susceptible cultivars A57 and HG70 (figure 2c). These patterns were consistent with the inoculation-associated community changes and higher post-inoculation diversity in resistant cultivars reported in the source study.

To examine the taxa underlying these resistance-associated differences, MicroBioAgent compared each resistant cultivar with the pooled susceptible cultivars using MaAsLin 3. In the 16S profiles, 6–15 genera were enriched in both HG64 and HF12 within each sampling condition ($P < 0.05$ and positive coefficients, Supplementary Figure S1), but the shared taxa varied across conditions. The matched shotgun data provided species-level comparisons of the markers highlighted in the source study. MetaPhlAn3 profiles showed enrichment of the *Pseudomonas putida* group in both the resistant parent HG64 ($P = 0.013$) and the resistant hybrid HF12 ($P = 0.009$) at T1. The other reported marker, *Sphingopyxis unclassified*, was not identified in the MetaPhlAn3 profiles. Following investigator review, MicroBioAgent repeated the shotgun profiling using MetaPhlAn2, the version used in the original study, and identified both *Pseudomonas putida* and *Sphingopyxis unclassified* as enriched in resistant cultivars, recovering both reported taxonomic associations.

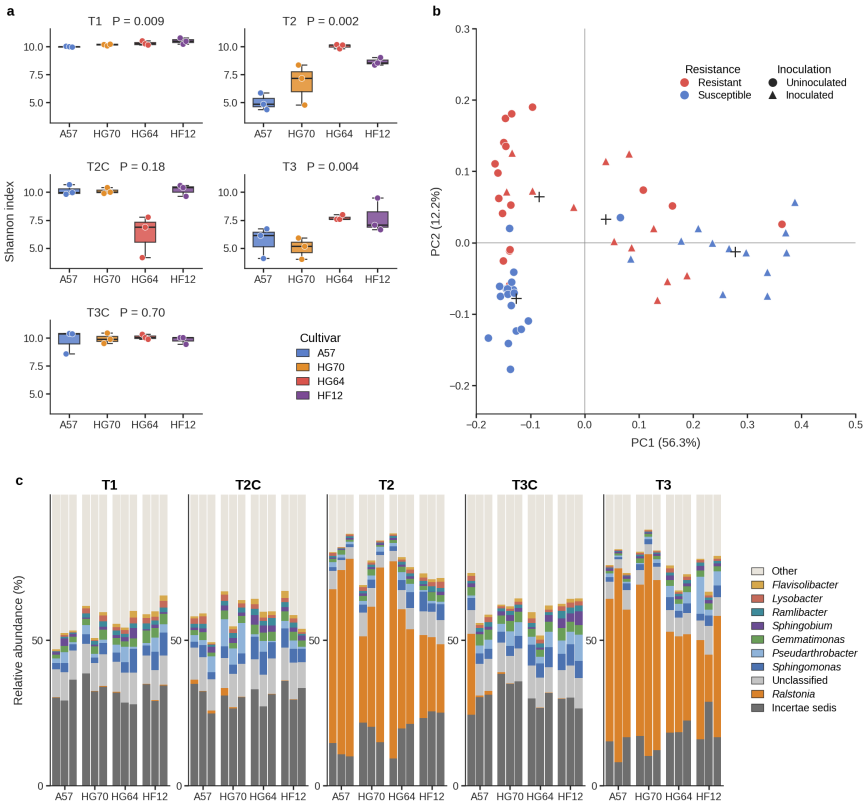


Figure 2: Tomato rhizosphere diversity and community composition. (a) Shannon diversity across cultivars, sampling time points, and inoculation conditions. (b) Weighted-UniFrac PCoA, coloured by resistance phenotype and shaped by inoculation treatment. (c) Genus-level composition across individual samples. Each cultivar-time-point combination contained three biological replicates.

2.2.2 A colorectal cancer faecal metagenomic biomarker study

The colorectal cancer (CRC) case examined disease-associated changes in microbial composition and the predictive value of microbial functional profiles. The dataset comprised 128 faecal metagenomes from the Hong Kong C1 cohort reported by Yu *et al.*, including 74 patients with CRC and 54 controls [13]. MicroBioAgent performed species profiles with MetaPhlAn3 and gene-family and KEGG Orthology (KO) abundance profiles with HUMAnN3 for downstream association analysis and classification (figure 3).

PCoA based on Bray–Curtis dissimilarities calculated from MetaPhlAn3 species-level relative abundance profiles showed no clear separation among controls, early-stage CRC cases ($n = 36$) and late-stage CRC cases ($n = 38$). PERMANOVA nevertheless detected differences in community composition associated with CRC status ($F = 2.32$, $P = 0.002$) and disease stage ($F = 1.65$, $P = 0.009$). Separate tests for sex, age and BMI were not significant. To identify the species associated with these community differences, MicroBioAgent compared species abundances between cases and controls, identifying eight CRC-enriched and two control-enriched taxa after Benjamini–Hochberg correction ($q < 0.05$). Four CRC-enriched species, *Parvimonas micra*, *Peptostreptococcus stomatis*, *Fusobacterium nucleatum* and *Solobacterium moorei*, together with control-enriched *Eubacterium ventriosum*, showed the same directions of association as reported in the original study. Among these ten differentially abundant species, Spearman co-occurrence analysis identified 16 positive correlations ($|\rho| \geq 0.4$, BH-adjusted $q < 0.05$). *Parvimonas micra* had the most connections, including positive correlations with *Fusobacterium nucleatum* ($\rho = 0.49$) and *Solobacterium moorei* ($\rho = 0.53$).

To assess whether microbial functional profiles could distinguish CRC cases from controls, a panel of 15 gene families and a separate panel of 15 KOs were selected from differentially abundant features using mRMR. Each panel was used to train a logistic-regression classifier evaluated by nested fivefold cross-validation within the C1 cohort. The gene-family panel achieved a pooled AUROC of 0.902, compared with 0.632 for the KO panel, indicating stronger discrimination using gene-family features in this cohort.

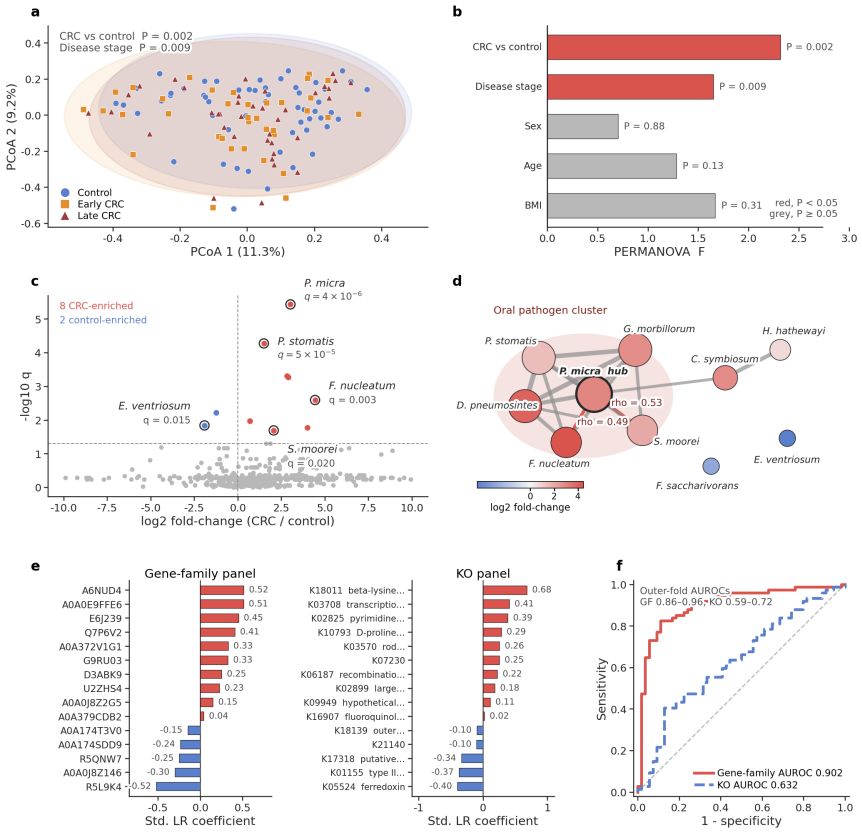


Figure 3: Reference-based analysis of CRC faecal metagenomes. Species-level beta diversity, disease-associated taxa, co-occurrence of the differentially abundant species, functional features, and nested cross-validation of gene-family and KO classifiers are shown.

2.2.3 A polyethylene-enrichment genome-resolved metagenomic study

We next evaluated whether MicroBioAgent could identify the predominant microorganisms in soil-derived enrichment cultures and characterize their genetic potential for degrading alkanes and their oxidized derivatives. The dataset comprised six shotgun metagenomes from two soil inocula (S1 and S2), each enriched on low-molecular-weight polyethylene (LMWPE), C30 or C40 [14]. MicroBioAgent assembled and binned these metagenomes to generate MAGs for taxonomic and functional characterization. Assembly and quality filtering yielded 153 high- or medium-quality MAGs (100 high quality and 53 medium quality). Dereplication at 95% average nucleotide identity retained 75 representatives (50 high quality and 25 medium quality, figure 4a).

MicroBioAgent used the MAG abundance profiles to examine which organisms predominated under each enrichment condition. *Acinetobacter guillouiae* dominated S1 LMWPE and S1 C30, accounting for 64% and 67% of the reported CoverM abundance, respectively. S1 C40 instead contained two dominant *Aquipseudomonas* species, sp002980155 (42%) and sp036884845 (23%, figure 4b,c). The S2 enrichments showed a different composition: *Aquipseudomonas* sp003050835 was most abundant in S2 LMWPE (42%), whereas *Mesorhizobium* and *Aquipseudomonas* species were prominent in S2 C30 and S2 C40. These descriptive differences varied with both inoculum and substrate, with each condition represented by a single metagenome. GTDB R232 assigned the former *Pseudomonas* MAGs to *Aquipseudomonas*, allowing the focal taxa to be related to those discussed in the source study despite the change in nomenclature.

To examine the degradation-associated functions carried by these organisms, MicroBioAgent annotated representative MAGs from ten focal taxa with eggNOG. Eight contained genes assigned to all three categories examined: alkane terminal oxidation, degradation of 2-ketones or oxidized derivatives, and β-oxidation (figure 4d). The dominant *Acinetobacter guillouiae* representative contained 4, 10 and 7 annotated genes in these categories, respectively. An *Aquipseudomonas* sp. representative had the highest counts for alkane terminal oxidation (11) and 2-ketone or oxidized-derivative degradation (19). The distribution was not uniform across the focal genomes: two lacked annotated alkane-terminal-oxidation genes, and one of these also lacked annotated M00087 β-oxidation genes. These annotations support the presence of genetic potential for processing alkanes and their oxidized derivatives, the substrates central to the source study. The gene counts do not establish pathway activity or degradation of the polyethylene polymer.

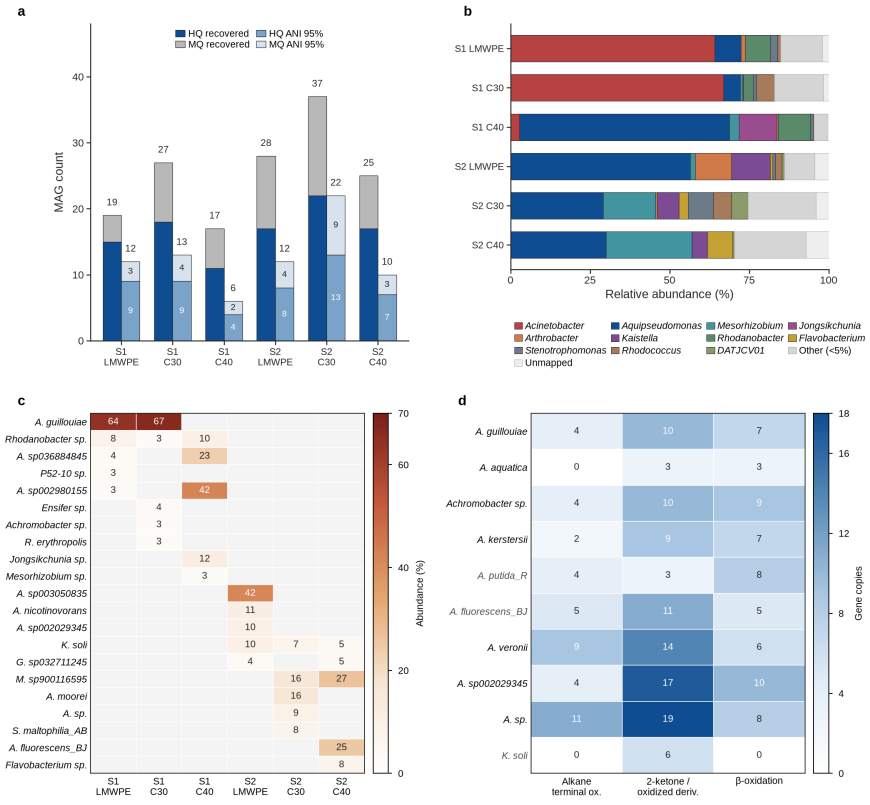


Figure 4: Assembly-based analysis of enrichment metagenomes. (a) Numbers of high- and medium-quality MAGs before and after 95% ANI dereplication. (b) Genus-level MAG composition. (c) Species among the five most abundant MAGs in at least one sample. (d) eggNOG-annotated gene counts in focal MAGs for alkane oxidation, 2-ketone or oxidized-derivative degradation, and β-oxidation.

2.3 Robustness across models and prompt granularity

Robustness will be evaluated on frozen Case 1 inputs with the complete system under three models, three prompt granularities, and three independent sessions per cell (27 sessions). The primary outcome is artifact-level completion of the common scientific goals: data characterization, 16S alpha and beta diversity, taxonomic and differential-abundance analysis, and shotgun taxonomic and differential-abundance analysis. Method-specific adherence is scored only for the detailed prompt. Shannon agreement, distance-matrix Mantel tests, and Procrustes comparisons of ordinations provide secondary repeatability measures.

Table 2: Planned reporting table for robustness across models and prompt granularity. Each cell summarizes three independent sessions. Values will be inserted after the 27-session experiment is complete.

Model	Prompt	Completed sessions	Artifact score mean / 5	Shannon agreement r	Mantel agreement r	Procrustes disparity
Model A	Detailed	pending	pending	pending	pending	pending
	Medium	pending	pending	pending	pending	pending
	Simple	pending	pending	pending	pending	pending
Model B	Detailed	pending	pending	pending	pending	pending
	Medium	pending	pending	pending	pending	pending
	Simple	pending	pending	pending	pending	pending
Model C	Detailed	pending	pending	pending	pending	pending
	Medium	pending	pending	pending	pending	pending
	Simple	pending	pending	pending	pending	pending

Notes: Completed sessions are reported as $n/3$. The artifact score sums five prespecified analysis domains, each scored 0, 0.5, or 1. Agreement statistics summarize pairwise comparisons among repeats within a model–prompt cell. Mantel results are reported separately by distance metric in the Supplementary Information, the main-table value summarizes metrics shared by all three sessions. Lower Procrustes disparity denotes closer agreement.

2.4 Evaluating component contributions and asynchronous guidance

2.4.1 Component contributions to end-to-end analysis

End-to-end component ablation will compare the complete system with conditions lacking shipped skills, coding and long-execution guidance, or the session-runtime extension. Each condition uses the same detailed Case 1 prompt, frozen inputs, one model, and three independent sessions without mid-run user feedback. The primary outcomes are end-to-end task success, artifact completion across nine requested outputs, adherence to the eight prespecified methods, and total token use. Task completion is additionally

resolved by output type to identify which parts of the requested analysis are omitted under each ablation. Raw token use will be interpreted together with completion because a stalled session can consume fewer tokens by performing less analysis.

Table 3: Planned reporting table for end-to-end component ablation. Each condition contains three independent sessions using the same model, detailed prompt, and frozen inputs.

A. Core ablation outcomes

Condition	Task completion rate (%)	Method adherence rate (%)	Median total tokens
Full	pending	pending	pending
−Runtime extension	pending	pending	pending
−Skills	pending	pending	pending
−Exec guidance	pending	pending	pending

B. Completion of individual requested tasks

Task	Full	−Runtime extension	−Skills	−Exec guidance
16S α diversity	pending	pending	pending	pending
16S β diversity	pending	pending	pending	pending
16S taxonomic table	pending	pending	pending	pending
16S differential abundance	pending	pending	pending	pending
MetaPhlAn table	pending	pending	pending	pending
Shotgun differential abundance	pending	pending	pending	pending
Figures	pending	pending	pending	pending
Scripts	pending	pending	pending	pending
Report	pending	pending	pending	pending

Notes: Full denotes the complete system, −Runtime extension uses the standard session runtime, −Skills removes the microbiome domain skills, and −Exec guidance removes the coding and long-execution instructions. Each task-specific cell in panel B reports the number of sessions that produced a valid result for that task ($n/3$). Task completion rate is the percentage of the nine task items completed, averaged across the three sessions in a condition. Method adherence rate is the percentage of the eight methods named in the detailed prompt: sequencing-depth normalization, Shannon diversity, weighted UniFrac, unweighted UniFrac, Bray–Curtis, weighted-UniFrac PCoA, PERMANOVA, and PERMDISP that were used as requested, averaged across the three sessions. Median total tokens use the same accounting source in all conditions. Individual-session and pre-failure token counts are reported in the Supplementary Information.

2.4.2 Runtime ablation for analysis continuity

We tested asynchronous guidance with a real, minutes-scale Case 1 QIIME 2 taxonomy-classification task. After the command was confirmed running, the user instructed the agent to replace a pooled resistant-versus-susceptible comparison with separate T1, T2C, and T2 comparisons and to add a significant-genera heatmap. No subsequent “continue” message was sent.

With the extended runtime, the agent yielded while the command continued, updated the pending plan during execution, and resumed automatically after command completion in all three sessions (table 4). With the standard runtime in the tested configuration, neither plan revision during active computation nor automatic session resumption after tool exit was observed in any of the three sessions. These observations support plan revision and automatic continuation for the tested command and runtime configuration. The two endpoints assess session interaction. They do not separately establish that every requested change was correctly reflected in the final analytical outputs.

Table 4: Runtime-extension ablation assessing asynchronous human guidance during active computation. Each runtime was evaluated in three independent sessions.

System	Plan updated during execution	Automatic continuation after tool exit
Extended runtime	(3/3)	(3/3)
Standard runtime	(0/3)	(0/3)

Notes: The two conditions differ only in the session-runtime extension described in section 4.3. Plan updated during execution requires the agent to yield while the tool continues in the background and to incorporate the user's message into the pending plan before tool completion. Automatic continuation requires tool exit to reactivate the session without another user message. Results are reported as successful sessions out of three.

3 Discussion

MicroBioAgent connects biological objectives to established microbiome methods through an analysis process that researchers can inspect and revise. The three reanalyses illustrate complementary uses of this approach: relating amplicon and shotgun profiles to resistance phenotypes, extending disease profiles into functional and predictive analyses, and linking taxonomically assigned MAGs to metabolic potential. Their common contribution is the composition of existing methods around a study question. The underlying algorithms and reference resources remain responsible for the primary biological estimates.

Organizing domain knowledge at the level of individual procedures provides a practical basis for this composition. Method requirements and expected products remain explicit while their arrangement can vary between studies. The case studies also show why interpretation must retain the context of these choices. A nominal association may not survive multiple-testing correction, a taxon may be represented differently across reference databases, and a predicted gene repertoire does not establish metabolic activity. Maintaining commands, parameters, and outputs within the analysis context supports review of such distinctions, although their correct interpretation

still requires scientific judgment.

The current evidence establishes feasibility across the selected cases. The planned robustness and component-ablation experiments have not yet established aggregate completion rates, model dependence, or the contribution of each component to analytical performance. Comparisons with a general agent given the same software and resources would further quantify the value of the integrated system. Agreement between repeated analyses should be assessed alongside method correctness and independently checked results, since consistent outputs can still be incorrect. User studies are also needed to determine whether the conversational interface reduces training, manual intervention, or time spent organizing an analysis.

Human confirmation of the initial plan and expert review of analytical choices remain part of the operating model. Procedure-specific output checks do not guarantee statistical validity or prevent unsupported interpretations. Local execution retains the compute, software-maintenance, and reference-database requirements of the underlying tools. Workflow managers remain well suited to repeated execution of a specified analysis at scale. MicroBioAgent focuses on defining and revising study-specific analyses through interaction.

Continuity during prolonged computation supports this operating model. The targeted runtime experiment demonstrated plan revision during an active command and automatic continuation after tool completion in the tested configuration. These observations establish a limited execution capability. reliability across longer jobs and more complex revisions, and the fidelity of final outputs to revised instructions, require further evaluation.

MicroBioAgent organizes established microbiome methods into study-specific analyses guided by biological questions. The three published-data reanalyses demonstrate paths through amplicon profiling, reference-based metagenomics, and MAG construction with taxonomic and functional characterization. By connecting modular domain procedures, local computational resources, and intermediate results within a common analysis context, the system provides a basis for question-driven analysis across microbiome workflows.

4 Methods

4.1 MicroBioAgent system overview

MicroBioAgent is an agent-based system for study-specific microbiome analysis, implemented on the OpenClaw runtime. It maintains a persistent analysis context across a conversation, linking the user's research objective and input data with the selected methods, intermediate products, and pending decisions. Before computation begins, the agent states an inspectable plan. It then selects modular analytical procedures according to the study objective,

available data, and verified intermediate results rather than requiring the entire study to conform to a predefined workflow (figure 1).

The system separates analytical knowledge from execution guidance. Domain procedures describe when a microbiome method is applicable, the inputs and resources it requires, the outputs it should produce, and the checks relevant to those outputs. Agent-facing execution instructions govern how procedures are run and how long-running commands release and later resume the conversation. A failed procedure is reported together with any downstream work that depends on it, unavailable results are not replaced by model-generated values.

4.2 Skill-guided analysis and configured local execution

MicroBioAgent combines a tool-use specification with step-level microbiome procedures. The tool-use instructions (`TOOLS.md`) direct the agent to the relevant domain procedure and associate it with a configured software environment and, where required, a reference resource. The domain procedures encode method-specific requirements without prescribing a single sequence for an entire study. Consequently, a statistical model or targeted annotation can be added to an existing profile when the scientific question requires it, without defining a separate end-to-end template.

Bioinformatics commands are executed in user-provided host or container environments. Software environments and reference paths are configured before analysis. If a required environment or database is unavailable, the agent reports the missing dependency rather than inferring an unverified path or silently substituting a resource. Compute-intensive processing of raw reads therefore occurs in the user-controlled environment and does not require their submission to an external microbiome analysis service. Optional online annotation procedures may transmit the query information required by the corresponding public service.

After an analytical operation finishes, the agent inspects its exit status and expected artifacts before using those products in subsequent work. The depth of this inspection is procedure-specific. It is an agent-guided check of expected outputs rather than a universal validation engine. The implemented analytical scope is summarized in table 1. Supervised modeling in the colorectal-cancer case was composed from the resulting abundance tables and general statistical software rather than exposed as a fixed classifier workflow.

4.3 Long-running analysis execution and continuation

A separate execution-guidance layer (`CODING.md`) instructs the agent to complete and inspect one principal analytical operation before advancing to dependent work. When a command remains active after an initial observation

period, the command continues while the conversational turn is yielded. The agent does not use conversational polling as a substitute for command completion.

Two events can subsequently reactivate the analysis. A user message allows the agent to revise pending work while the command continues. If the command is still active after that revision, the conversation is yielded again. A separate completion event reactivates the session when the command exits, at which point the outputs are inspected and work continues under the current plan.

We implemented this continuation mechanism through a version-pinned extension to the OpenClaw session runtime. It builds on existing facilities for background execution and yielding a session. The extension adjusts completion-event delivery, resumption of the associated session, and runtime tool guidance for yielding control during active computation. It also handles completion events that arrive while a conversational turn is still active. The execution instructions specify how the agent uses these facilities. Domain procedures guide the analytical work performed after resumption. Asynchronous guidance therefore depends on the interaction of runtime behavior and agent instructions. For reproducibility, the extension is distributed as a version-pinned OpenClaw patch pack with the complete instruction text and application instructions. The target runtime version and commit are recorded in the accompanying manifest.

4.4 Scope of the related-system comparison

Table 5 summarizes reported functional scope in related bioinformatics agents [6–11, 15–17]. The comparison concerns documented capabilities, not comparative accuracy or performance under a shared evaluation. Unconfirmed capabilities are marked as unestablished rather than absent.

4.5 Case study design

We selected three published datasets to exercise complementary analytical capabilities on plant, human, and environmental microbiomes (Supplementary Table S1). The cases were designed as capability-oriented reanalyses rather than exact computational reproductions. Software releases, reference databases, and selected statistical procedures could therefore differ from the source publications. Before examining the reanalysis output, we identified the biological comparisons and taxa or functions emphasized by each source study. Evaluation considered whether the requested analysis was completed, whether its artifacts were usable, and whether those prespecified signals were recovered. Numerical identity with the source publication was not required.

Table 5: Reported capabilities of bioinformatics agents relevant to end-to-end microbiome analysis. The three microbiome columns specifically concern analysis beginning with sequencing reads.

	Agent operation		Microbiome analysis from reads			Deployment
Agent	Autonomous workflow execution	Human-in-the-loop	16S amplicon	Reference-based shotgun	Assembly / MAG	On-premises analysis
BIA	✓	—	—	—	—	✓
BioChatter	—	✓	—	—	—	✓
AutoBA	✓	—	—	—	—	✓
Biomni (open-source)	✓	—	—	—	—	✓
BioAgents	—	—	—	—	—	—
M3A	✓	✓	—	—	—	✓
BioMedAgent	✓	—	—	—	—	✓
BioMaster	✓	—	—	✓	—	✓
MetaClaw	✓	—	✓	✓	—	✓(partial)
MicroBioAgent	✓	✓	✓	✓	✓	✓

Notes: ✓, reported support, —, support not established from the reviewed sources, which does not demonstrate inability. Biomni denotes the open-source release. This scope comparison is not a performance benchmark.

4.5.1 Case 1 — Tomato rhizosphere 16S and metagenomic analysis

The tomato dataset comprised 60 V3–V4 16S samples arranged as four cultivars, five sampling conditions, and three biological replicates [12]. A57 and HG70 were treated as susceptible cultivars, whereas HG64 and HF12 were treated as resistant. The sampling conditions were T1 (day 0), T2C and T2 (day 5, uninoculated and inoculated), and T3C and T3 (day 10, uninoculated and inoculated). Thirty-six matched shotgun libraries covered T1, T2C, and T2.

Amplicon sequence variants were obtained by DADA2 denoising and assigned taxonomy using the configured QIIME 2 classifier. Alpha diversity was calculated from an ASV table rarefied to 56,726 reads per sample. Shannon values were summarized by cultivar, time point, and inoculation status. Pooled resistant and susceptible groups were compared within each time point using two-sided Mann–Whitney tests. Community dissimilarity was evaluated using weighted and unweighted UniFrac and Bray–Curtis distances. Weighted-UniFrac principal-coordinate analysis was used for visualization, and group effects were assessed by PERMANOVA with PERMDISP used to evaluate homogeneity of multivariate dispersion.

Genus-level differential abundance was evaluated separately at each sampling condition. MaAsLin 3 models used total-sum scaling followed by logarithmic transformation and contrasted each resistant cultivar with the pooled susceptible cultivars. A genus was classified as shared nominal enrichment

when its coefficient was positive and $P < 0.05$ in both the HG64 and HF12 contrasts. Benjamini–Hochberg-adjusted results were retained to distinguish nominal intersections from false-discovery-rate-supported associations.

Matched shotgun libraries were profiled at species level. MetaPhlAn3 was the primary reference-based analysis, and MetaPhlAn2 was used as a database-version sensitivity analysis for the two taxa emphasized by the source study, *Pseudomonas putida* and *Sphingopyxis unclassified*. Detection and between-phenotype abundance results were reported separately for the two database generations rather than selecting one profile as the definitive answer.

4.5.2 Case 2 — Colorectal cancer faecal metagenomic analysis

The colorectal-cancer analysis used the C1 cohort reported by Yu *et al.*, comprising 74 CRC cases and 54 controls [13]. Among the cases, 36 were stage I–II and 38 were stage III–IV. Species relative abundances were obtained with MetaPhlAn3, and HUMAnN3 was used to quantify gene families and KEGG Orthology groups.

Species-level Bray–Curtis dissimilarities were analysed by principal coordinates analysis. PERMANOVA tested disease status, cancer stage, sex, age, and body-mass index. Species, gene families, and KO groups were compared between CRC and control samples using two-sided Wilcoxon rank-sum tests with Benjamini–Hochberg correction. For the differentially abundant species, pairwise Spearman correlations were corrected for multiple testing. the displayed co-occurrence network retained positive associations with $\rho \geq 0.4$ and adjusted $P < 0.05$.

Predictive analyses were performed separately on gene-family and KO abundance tables. Fifteen features were selected using minimum-redundancy maximum-relevance and entered into L2-regularized logistic regression. Model performance was assessed by nested fivefold cross-validation, with feature selection and model fitting confined to the corresponding training data. Pooled areas under the receiver-operating-characteristic curve were calculated from out-of-fold predictions. These analyses assessed execution of a taxonomic–functional–predictive study design within this cohort and were not treated as external clinical validation.

4.5.3 Case 3 — Polyethylene-enrichment genome-resolved metagenomic analysis

The genome-resolved analysis used six shotgun metagenomes from two soil inocula enriched on low-molecular-weight polyethylene, C30, or C40 [14]. MAGs were constructed through assembly and binning in the configured genome-resolved analysis environment. MAGs with completeness $\geq 90\%$ and contamination $< 5\%$ were classified as high quality. Those with completeness

$\geq 50\%$ and contamination $< 10\%$ were classified as medium quality. MAGs were dereplicated at 95% average nucleotide identity, and taxonomy was assigned using GTDB release R232.

Reads were mapped to the dereplicated MAG set, and CoverM-derived relative abundances were summarized at genus and species levels. Because each inoculum–substrate combination was represented by one metagenome, these profiles were interpreted descriptively and MAGs were not treated as independent biological replicates.

One representative MAG was selected for each focal taxon and annotated with eggNOG. Predicted coding sequences were counted against rules representing alkane terminal oxidation, degradation of 2-ketone or other oxidized derivatives, and the KEGG M00087 β-oxidation module. Counts therefore represent predicted genomic coding potential, not expression measurements or direct evidence of substrate degradation. The complete gene-to-module rules are reserved for the Supplementary Methods.

4.6 Robustness and ablation experiments

4.6.1 Robustness across models and prompt granularity

Robustness will be evaluated with frozen Case 1 inputs in a $3 \times 3 \times 3$ design comprising three language models, three prompt granularities, and three independent sessions per cell (27 sessions). The complete system, computational environment, skills, and staged data will be held fixed. Detailed prompts specify the metrics, statistical tests, and figure types, medium prompts specify the same analytical goals but leave method selection to the agent, simple prompts state only the dataset and biological question. Each replicate will begin in a new conversation, and methodological coaching will not be provided after the plan is confirmed.

The 16S inputs comprise a post-DADA2 ASV table, representative sequences, rooted tree, denoising statistics, and metadata. Shotgun inputs comprise the 36 matched libraries after KneadData processing and random subsampling to 20% of reads with a fixed seed of 42. Previously generated diversity, MetaPhlAn, statistical, and figure outputs are excluded.

Completion will be scored from usable artifacts for data characterization, 16S alpha diversity, 16S beta diversity, 16S taxonomy and differential abundance, and shotgun taxonomy and differential abundance. Each domain will receive 0, 0.5, or 1. Method adherence will be scored separately for detailed prompts. Agreement among the three repeats of a cell will be assessed using pairwise correlation of Shannon values, Mantel correlation of shared distance metrics, and Procrustes comparison of ordinations. With three repeats per cell, these comparisons will be treated as an initial repeatability screen.

4.6.2 Component-ablation design

The component ablation will use the detailed robustness prompt, the same frozen inputs, and one fixed model. Four conditions will be evaluated in three independent sessions each: the complete system, the standard session runtime without the extension, removal of the microbiome domain procedures, and removal of the coding and long-execution guidance. No feedback will be introduced while a session is running. All other workspace instructions, software resources, and reference resources will be held constant wherever they are not the component under test.

Task completion will be scored using nine binary items: 16S alpha diversity, 16S beta diversity, a 16S taxonomic table, 16S differential abundance, a MetaPhlAn table, shotgun differential abundance, figures, analysis scripts, and an analysis report. An item will be counted only when the corresponding artifact exists and is usable. Method adherence will be scored against eight requirements in the detailed prompt: sequencing-depth normalization, Shannon diversity, weighted UniFrac, unweighted UniFrac, Bray–Curtis, weighted-UniFrac PCoA, PERMANOVA, and PERMDISP. Task and method rates will be calculated per session and averaged within condition. Total token use will be obtained from the same accounting source for every session and interpreted together with completion because an early failure can consume fewer tokens by performing less work.

4.6.3 Runtime-extension experiment

Asynchronous intervention was evaluated with a Case 1 QIIME 2 taxonomy-classification command selected in a pilot to run for several minutes without artificial delay. The extended runtime and the standard runtime at the same OpenClaw version were each evaluated in three new sessions. The two conditions differed only in the session-runtime extension. All other configuration and workspace instructions were held fixed. While classification was confirmed to be running, the same user instruction requested replacement of a pooled resistant-versus-susceptible comparison with separate T1, T2C, and T2 comparisons and addition of a significant-genera heatmap. No subsequent user message was sent.

A plan-update success required the message to be processed and the pending plan to be revised before the classification command exited. Success for automatic continuation required command exit to reactivate the same session within five minutes without another user message. A run in which classification ended before the intervention was delivered did not test in-run feedback and was repeated with its exclusion reason retained.

Data availability

Raw sequencing data are available from the public repositories associated with the three source studies. Accession identifiers and sample-level mapping will be listed in the Supplementary Information. Processed abundance tables, metadata, numerical data underlying the figures, MAG annotations, and session-level robustness and ablation scores will be deposited with the article archive. Raw sequencing files and large reference databases will not be redistributed. Their sources and releases will be documented.

Code availability

MicroBioAgent source code, domain procedures, agent instructions, analysis prompts, environment specifications, and the session-runtime extension (distributed as a version-pinned OpenClaw patch pack) are available in the project repository. Scripts used to generate the reported statistical results and figures will be included in the accompanying archive.

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Supplementary Information

Table S1: Published datasets, sequencing-library counts, analysis tasks, and evaluation targets for the three MicroBioAgent case studies.

Case	Sequencing libraries	Analysis route	Primary evaluation target
Tomato rhizosphere	60 16S; 36 matched shotgun	16S diversity, taxonomy, differential abundance; MetaPhlAn profiling	Recovery of inoculation- and resistance-associated community signals
CRC faecal metagenomes	128 shotgun	MetaPhlAn3, HUMAnN3, differential analysis, nested-CV classification	Taxonomic and functional case–control analysis from reads
Enrichment metagenomes	6 shotgun	MAG construction, dereplication, taxonomic assignment, abundance, eggNOG annotation	Genome-resolved taxonomic and functional characterization

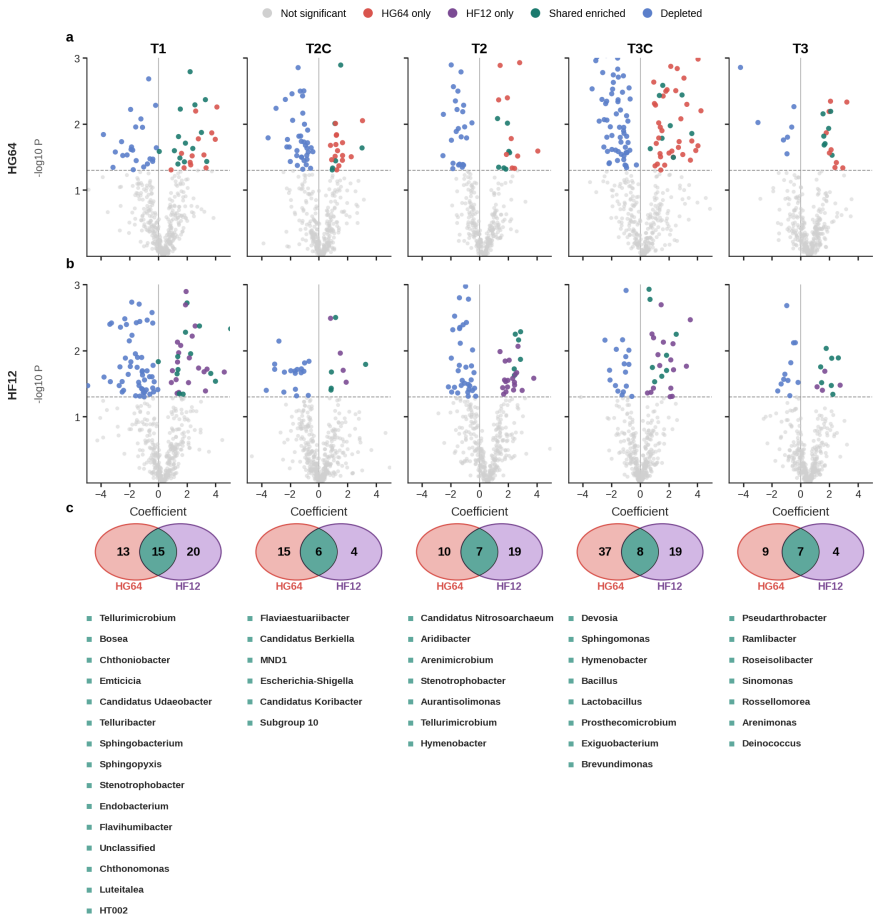


Figure S1: Differentially enriched genera in the resistant tomato parent HG64 and progeny HF12. MaAsLin 3 compared each resistant cultivar with pooled susceptible cultivars at each sampling point. Shared enrichment denotes a positive coefficient and nominal $P < 0.05$ in both contrasts.